The Link Function Problem in Psychological Interaction Testing

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Abstract

Psychological interaction claims are claims about moderation, boundary conditions, and heterogeneous effects. We argue that such claims depend on the scale on which additivity is assumed. In generalized linear models, that scale is defined by the link function. Choosing an appropriate outcome family is therefore necessary but not sufficient: researchers must also choose, justify, and probe a plausible link. We first report a preregistered descriptive review of recent psychological articles, showing that interaction tests are common but link-function choices are often implicit, especially for bounded, discrete, ordinal, and sum-score outcomes. We then develop a link-centered framework that distinguishes interaction on the linear predictor scale from interaction on the observed response scale, and separates wrong-family, wrong-link, and wrong-metric problems. Worked examples focus on forced-choice accuracy with a non-zero chance floor, sum scores as bounded and discrete outcomes, and within-family link choices such as logit versus probit. Simulations quantify when wrong or inadequate links can generate pseudo-interactions, and one diagnostic worked example illustrates why model checks are useful but not sufficient. We close with a practical workflow for link-aware interaction testing in psychological research.

Keywords: interactions, moderation, generalized linear models, link functions, model misspecification

The Link Function Problem in Psychological Interaction Testing

Psychological theories often ask whether the effect of one variable depends on another. Stress may impair performance more strongly among people with high anxiety. A treatment may help some participants more than others. A developmental difference may be large at one age and small at another. These are interaction claims: claims about moderation, boundary conditions, and heterogeneous effects. They are central to psychological theory because they move beyond the question of whether an effect exists and toward the more substantive question of when, for whom, and under which conditions it occurs (Domingue et al., 2024; Rohrer & Arslan, 2021).

Statistically, many interaction claims can be understood as claims about a difference between differences. Does the treatment-control difference change across groups? Does the age-related difference differ between populations? Does the effect of task difficulty vary as anxiety increases? This formulation is intuitive, but it hides a crucial assumption. A difference is always defined on some scale. An interaction is a difference between differences, but differences are not scale-free.

This matters because psychological outcomes are often observed on scales that are not simple linear reflections of the quantities researchers care about. Equal intervals on an underlying psychological or model-based scale may correspond to unequal intervals on the observed response scale. Conversely, equal observed differences may correspond to unequal differences on the underlying scale. This is obvious for probabilities, counts, response times, bounded ratings, and sum scores, but the same logic applies more broadly whenever the observed response is a nonlinear, bounded, discrete, or otherwise imperfect representation of the process of interest.

Thus, before asking whether two differences differ, researchers must clarify the scale on which those differences are defined. A pattern may look like moderation on the observed response scale but not on a transformed scale, or it may look additive on the linear predictor scale but nonadditive after being mapped back to observed responses. This does not mean that all interactions are artifacts or that scale choices make interaction claims arbitrary. It means that interaction claims require an explicit statement of the scale on which additivity is assumed and on which departures from additivity are interpreted.

This point has a long history. Loftus argued that interactions involving response probabilities are meaningful only relative to an assumed mapping between the observed response and the theoretical component of interest (Loftus, 1978). Cox reviewed interaction as a statistical concept tied to model formulation, transformation, and interpretation (Cox, 1984). Wagenmakers and colleagues later emphasized the distinction between removable and nonremovable interactions, showing that some interactions are tied to the measurement scale and therefore do not support unambiguous conclusions about latent psychological processes (Wagenmakers et al., 2012). Lubinski and Humphreys illustrated how apparently substantive moderator effects can arise from model specification choices rather than genuine psychological synergy (Lubinski & Humphreys, 1990). More recently, Rohrer and Arslan argued that interaction tests often provide precise answers to vague questions, in part because researchers do not always specify the scale or estimand their interaction claim concerns (Rohrer & Arslan, 2021).

A related literature has sharpened the problem for modern psychological data. Domingue and colleagues showed that misspecifying the outcome model can produce bias and false discoveries in interaction analyses, especially when outcomes are binary, count, censored, or noninterval (Domingue et al., 2024). McCabe and colleagues showed that, in generalized linear models (GLMs) for probabilities and counts, product-term coefficients are not generally equal to interaction effects on the response scale (McCabe et al., 2022). Work on nonlinear probability models reaches the same practical conclusion: interaction effects should often be evaluated as conditional marginal effects or response-scale contrasts, not as product coefficients alone (Ai & Norton, 2003; Mize, 2019). Rimpler and colleagues further emphasized that a theoretical claim of conditionality does not automatically imply that the correct statistical representation is a multiplicative product term, and that functional-form misspecification can generate misleading interactions (Rimpler et al., 2026). Together, these contributions make clear that interaction testing is not only a matter of adding the right-hand-side product term to a regression equation.

This paper focuses on one specific part of this broader problem: the role of the link function in GLM interaction testing. In the standard GLM architecture, the family and the link answer two different modeling questions (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972). The family answers: what kind of outcome am I modeling? It specifies the support and sampling process of the response, for example binary choices, counts, positive continuous times, or approximately Gaussian scores. The link answers: on which scale do I assume that effects add? It maps the conditional mean of the outcome to the linear predictor and therefore defines the scale on which a model without a product term is additive.

Choosing an appropriate family is therefore necessary, but it is not sufficient for interaction testing. Consider response times. A Gamma family may be plausible because response times are positive, continuous, and often right-skewed. In this respect, both `Gamma(link = "log")` and `Gamma(link = "identity")` may respect important features of the outcome. Yet they encode different no-interaction baselines. With a log link, additivity is assumed for log mean response times. On the observed scale, this corresponds to proportional or multiplicative change: a condition may increase response time by the same percentage across groups, while producing larger absolute millisecond differences when baseline times are longer. With an identity link, additivity is assumed directly in mean milliseconds: the same condition is expected to add the same number of milliseconds across groups.

This difference matters because an interaction is evaluated relative to the model's no-interaction baseline. If the data are additive on the log scale but the analyst fits a Gamma identity model, the response-scale curvature can be absorbed by a product term and interpreted as moderation. Conversely, if the theoretically relevant effect is additive in milliseconds, a log link may define the wrong baseline. Thus, a pseudo-interaction can emerge even when the family is plausible, because the link function imposes the scale on which "no interaction" is defined. Link misspecification can affect likelihood-based inference in GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019).

The same logic applies outside accuracy data. Response times provide a simple bridge case. A Gamma family may be plausible because response times are positive, continuous, and often right-skewed. In this respect, both `Gamma(link = "log")` and `Gamma(link = "identity")` may respect important features of the outcome. Yet they encode different no-interaction baselines. With a log link, additivity is assumed for log mean response times. On the observed scale, this corresponds to proportional or multiplicative change: a manipulation can have the same effect on the log scale while producing larger absolute millisecond differences when baseline times are longer. With an identity link, additivity is assumed directly in mean milliseconds: the same manipulation is expected to add the same number of milliseconds across groups.

Thus, a pseudo-interaction can emerge even when the family is plausible, because the link imposes the scale on which "no interaction" is defined. If the data are additive on the log scale but the analyst fits a Gamma identity model, the response-scale curvature can be absorbed by a product term and interpreted as moderation. Conversely, if the theoretically relevant claim concerns additive millisecond differences, a log link may define the wrong baseline. The Gamma example is not one of the central cases of this paper. It is a compact illustration of the broader principle developed below for forced-choice accuracy, sum scores, and within-family link choices such as logit versus probit.

The simple link map in Figure 2 visualizes this distinction for a binomial outcome.

Figure 3 uses one simulated 2-AFC accuracy dataset as a hook. Each participant completes 50 trials, and the outcome is the proportion correct. The predictors are age and group. The data-generating model includes an age effect and a group effect, but no age-by-group term on the chance-corrected 2-AFC logit scale, the scale on which the task has a .50 chance floor. The key point is that the same data can lead to different apparent conclusions about moderation because different models define different no-interaction baselines. A Gaussian identity model asks whether group differences change on the observed proportion scale. A standard binomial logit or probit model asks whether effects are additive on a link scale that maps the linear predictor to the full 0-to-1 probability range. A chance-corrected binomial link asks the same question after constraining the response range to the theoretically possible .50-to-1 interval. Thus, the figure is not a model-selection exercise, but a scale-of-null exercise: the interaction claim depends on which scale defines "no moderation" (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022).

The three panels fit the same data with three different models. The identity-link model treats proportions as approximately continuous outcomes and assumes additivity on the observed accuracy scale. The standard binomial-logit model respects the binomial sampling process and the 0-to-1 range of probabilities, but it assumes that performance can approach 0. The 2-AFC binomial-logit model respects an additional feature of the task: chance performance is .50, not 0. The figure is not meant to prove that one can obtain any result by changing links. It is meant to show that the scientific claim "age moderates the group difference" is incomplete unless the scale of additivity has been specified.

The 2-AFC example also shows why the issue is not simply "linear model versus GLM." A standard binomial model is often more appropriate than a Gaussian model for trial-level or aggregated accuracy data, but a standard binomial-logit link still maps the linear predictor to the full 0-to-1 probability range. In a 2-AFC task, however, the lower bound implied by random guessing is .50. A chance-corrected link can encode this constraint by mapping the linear predictor to the interval from chance to 1. This kind of link is familiar in psychophysics and item-response models with guessing parameters, but it is rarely treated as a routine part of GLM-based interaction testing in psychology. Related work on binary GLMs shows that misspecified links can bias likelihood-based inference, and that the problem extends to mixed models with clustered binary responses (Czado & Santner, 1992; Yu & Huang, 2019). Our point is not that every forced-choice analysis requires the same correction. Rather, when the scientific claim concerns moderation in a task with a known chance floor, the choice between a standard and a chance-corrected link becomes part of the interaction claim.

The importance of link choice also depends on the goal of the analysis. If the goal is purely predictive, researchers may reasonably prioritize predictive accuracy, calibration, generalizability, and decision usefulness. A model can be useful for prediction even when its parameters are not a faithful description of the psychological process. But if the goal is scientific inference about moderation, the link function carries substantive content. It defines the scale on which effects are additive or non-additive. In that setting, the question is not only whether the model predicts well, but whether the scale on which it defines the interaction matches the scale on which the scientific claim is being made. This is consistent with estimand-centered approaches that treat models as tools for generating substantively meaningful quantities rather than as collections of coefficients to be interpreted mechanically (Rohrer & Arel-Bundock, 2026).

Our contribution is therefore narrow. We do not claim that link functions have been ignored, nor that scale dependence is a new idea. Instead, we argue that psychology lacks an explicit, interaction-focused framework for choosing among plausible link functions and for quantifying how wrong or inadequate link choices can generate pseudo-interactions in realistic psychological settings. This framework is especially important for the case in which researchers have already moved beyond a simple linear model and chosen a broadly plausible GLM family, but have not examined whether the link function encodes the scale of additivity implied by their scientific question.

We develop this argument in three applied cases. The central case is forced-choice accuracy with a non-zero chance floor. Here, the problem is that standard binomial links respect the probability range but may fail to respect the task-implied lower bound. The second case is sum scores, which are common in psychology but often discrete, bounded, skewed, and only indirectly related to latent constructs. We do not argue that sum scores are invalid. We argue that interaction claims based on sum scores require scale-aware sensitivity analyses when additivity on the manifest sum-score scale is not clearly justified, especially because ordinal and metric analyses can diverge in ways that affect interactions (Liddell & Kruschke, 2018). The third case is within-family link choice, such as logit versus probit. These links are often practically similar, especially in the middle of the probability range, but their differences can matter for interaction claims near floors, ceilings, chance levels, or distributional tails.

We also discuss diagnostics, but with a limited claim. Link tests, residual diagnostics, calibration checks, and predictive checks can reveal important forms of misfit (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980). They should be part of responsible modeling practice. However, diagnostics do not by themselves certify the scientific validity of an interaction claim, and they may not distinguish a wrong link from a missing nonlinearity, a missing interaction, a wrong family, or a problematic measurement metric. For this reason, we treat diagnostics empirically in one focused worked example. In that example, we report several DHARMa residual checks separately rather than collapsing them into a single omnibus decision, because these checks are partly overlapping summaries of the same simulated residuals. The goal is descriptive: to examine which diagnostics, if any, detect the wrong-link problem in the same simulations where the wrong link can generate a false-positive interaction.

The rest of the paper proceeds as follows. First, we report a descriptive review of current practice in psychological interaction testing, focusing on how often interaction studies make link functions explicit and which outcome types are commonly analyzed. Second, we define the link function problem and distinguish interactions on the linear predictor scale from interactions on the observed response scale. Third, we examine forced-choice accuracy, sum scores, and within-family link choice as applied cases where link assumptions can matter for scientific moderation claims. Fourth, we use simulations to quantify when wrong or inadequate link choices generate pseudo-interactions. Fifth, we evaluate diagnostics in one selected scenario to show what they can and cannot detect. We close with a minimal workflow for link-aware interaction testing in psychological research.

Current Practice in Psychological Interaction Testing: A Preregistered Review

Review Rationale

Is this a technical corner case, or does it matter for current psychological practice? This question motivated the preregistered descriptive review reported in this section. The review was designed to answer two simple questions: how often do psychological articles deal with interactions, and whether the link-function problem is relevant for the kinds of outcomes that researchers actually analyze. We treat the review as descriptive evidence about reporting and model specification, not as an audit of whether individual articles are right or wrong. In most published reports, the information needed to evaluate the substantive status of a specific interaction is limited. Authors may not report the link function explicitly, may describe the model family without clarifying the scale of additivity, may analyze transformed or aggregated outcomes without fully specifying the measurement assumptions, or may target an observed-scale estimand that is defensible but not stated in those terms. For the same reason, we did not systematically classify reported interactions as removable or nonremovable. That distinction is conceptually important (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012), but it often cannot be recovered reliably from the published report alone, especially when the outcome model, link function, raw response scale, and theoretical scale of interest are only partly described. The purpose of the review is therefore narrower and more pragmatic. We ask whether interaction tests are common, whether family and link choices are usually explicit, and whether many interaction claims involve outcomes for which the scale of additivity is not trivial, such as probabilities, counts, ordinal responses, response times, and sum scores. If these choices are often implicit, then many moderation claims are difficult to interpret, not necessarily because they are false, but because readers cannot tell which no-interaction baseline was assumed. This is the empirical motivation for the rest of the paper.

This review is a central part of the present paper. The link-function problem would be less urgent if interaction tests were rare, or if researchers routinely justified the scale on which interactions were tested. Conversely, if interaction tests are common and the link function is often implicit, then the scale of additivity is often implicit as well. This would mean that many published moderation claims are difficult to evaluate, not necessarily because they are wrong, but because the scale on which they are made is underspecified. This focus on explicit model reporting is consistent with broader editorial recommendations that quantitative manuscripts should make analytic choices clear enough for readers to evaluate them (Mustillo et al., 2018).

Review Method

The review protocol was preregistered before the current manuscript was written. The protocol specified a descriptive review of recently published psychological research articles, with the primary objective of estimating how often empirical articles test interactions, how often they use or explicitly report non-identity link functions, how often identity-link analyses may be problematic given the outcome type, and what response-variable types are used in interaction analyses. The protocol also specified that the review was not intended as an exhaustive systematic review of the entire psychological literature, but as a targeted estimate of current practice in selected journals.

Articles were sampled from five journals selected to represent different areas of psychology: *Psychological Science*, *Journal of Experimental Psychology: General*, *Journal of Personality and Social Psychology*, *Developmental Science*, and *Psychological Medicine*. The protocol specified that articles would be retrieved from Scopus and screened for eligibility. Eligible articles were empirical articles presenting original psychological research. Reviews, meta-analyses, editorials, theoretical articles without empirical data, qualitative-only articles, replication studies, and psychometric validation studies were excluded. Articles were then coded in detail if they tested at least one interaction effect, either through statistical modeling or analysis of variance.

For each interaction-testing article, coders recorded whether at least one tested interaction used a non-identity link function, whether the link function was explicitly reported, whether at least one interaction was analyzed with an identity link when another link appeared more appropriate given the outcome, whether at least one such interaction was statistically significant or interpreted analogously, and which types of response variables were analyzed. The protocol specified descriptive analyses, including proportions and Wilson 95% confidence intervals for Boolean variables. The full review protocol, coding definitions, and extended descriptive tables are reported in the Supplement.

Review Results

The final dataset included 331 eligible empirical articles. Of these, 200 articles tested at least one interaction effect, corresponding to 60.4%, 95% CI [55.1, 65.5]. Among the 200 interaction-testing articles, 58 used at least one non-identity link function, 29%, 95% CI [23.2, 35.6]. Only 46 explicitly reported or otherwise identified the link function, 23%, 95% CI [17.7, 29.3].

Identity-link analyses were common in contexts where the outcome type made the scale of additivity worth justifying. Among the 200 interaction-testing articles, 144 were coded as including at least one potentially problematic identity-link analysis, 72%, 95% CI [65.4, 77.8]. Among these 144 cases, 124 reported at least one statistically significant interaction, 86.1%, 95% CI [79.5, 90.8]. Table 1 summarizes these results.

The response variables used in interaction analyses were diverse. Broad, non-exclusive categories included binary, accuracy, or proportion outcomes; sum scores and composite scores; ordinal, Likert, or rating outcomes; response times and durations; counts or error counts; neural or physiological measures; correlations or association indices; and difference or distance scores. This heterogeneity matters because the plausibility of an identity link depends on the outcome and on the scientific claim. For some continuous outcomes, additivity on the observed scale may be defensible. For bounded, discrete, ordinal, chance-constrained, or heavily transformed outcomes, however, additivity on the observed scale is an assumption that should usually be made explicit and probed. This is particularly relevant for ordinal outcomes, where treating ordered categories as metric can change effect estimates, error rates, and interaction conclusions (Bürkner & Vuorre, 2019; Liddell & Kruschke, 2018).

What the Review Shows, and What It Does Not Show

These results should be interpreted descriptively. The code “potentially problematic identity-link analysis” is not a verdict that a published interaction claim was false. In some cases, the identity scale may have been defensible. In others, the interaction claim may have been robust to alternative links, or the authors may have intentionally targeted an observed-scale estimand.

The review also did not code whether reported interactions were removable or nonremovable, because published reports often do not provide enough information to determine this reliably.

The review does show, however, that interaction testing is common and that the scale on which interactions are tested is often not made explicit. This is the empirical opening for the rest of the paper. If the link function defines the scale of additivity, then failing to report, justify, or probe the link function leaves a scientific moderation claim partly unspecified. The next section therefore develops the link-function problem directly: what the link function does, why choosing the family is not sufficient, and how product-term interactions in GLMs should be interpreted.

The Link Function Problem

The review shows that interaction tests are common and that the link function is often implicit. We now turn to the conceptual issue that makes this practice consequential. In a GLM, the link function is not only a computational device. In the standard GLM architecture, it is the part of the model that relates the conditional mean to the linear predictor (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972). It therefore defines the scale on which the predictors are assumed to combine additively. As a result, a product-term interaction in a GLM is always an interaction relative to a particular scale.

In a standard GLM, the expected outcome for observation i, denoted $\mu_i = E(Y_i)$, is related to a linear predictor through a link function:

$$g(\mu_i) = \eta_i.$$

For a model with two predictors, X_i and Z_i, and their product term, the linear predictor can be written as

$$\eta_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i.$$

The family specifies the outcome model, such as Gaussian, binomial, Poisson, or Gamma. The link function specifies how the mean of that outcome model is mapped onto the linear predictor. The coefficient β_3 is therefore a product-term interaction on the $g(\mu)$ scale. When

$\beta_3 = 0$, the model assumes additivity on the link scale. It does not necessarily assume additivity on the observed response scale.

What the link function does

The linear predictor, η, is the model's working scale. It is built by adding the intercept, main effects, product terms, and any other model terms, so in principle it can vary from minus infinity to plus infinity. Many psychological outcomes cannot. Response times must be positive. Counts cannot be negative. Probabilities must stay between 0 and 1. Forced-choice accuracy may have a floor above zero, for example .50 in a two-alternative task. Sum scores have a minimum and a maximum, often with discreteness and ceiling or floor effects.

The link function connects these two scales. In the notation $g(\mu) = \eta$, g maps the expected response to the linear predictor, and the inverse link maps η back to the expected response. Keeping predictions inside the admissible range is only the first role of the link. Its more important role for interactions is to define where effects are additive. Equal changes in η are treated as equal model-scale effects, but after the inverse link they may become unequal changes in milliseconds, counts, probabilities, accuracy, or sum-score units. Thus, changing the link changes the scale on which "no interaction" is defined.

This distinction is easy to miss because it disappears in ordinary linear models with an identity link. If $g(\mu) = \mu$, then additivity on the linear predictor scale is also additivity on the expected response scale. A one-unit change in X has the same interpretation in the model equation and on the observed outcome scale. For nonlinear links, this equivalence no longer holds. A logit link assumes additivity in log odds. A probit link assumes additivity on a latent-normal scale. A log link assumes additivity in log means, which corresponds to multiplicative change on the response scale. Each link defines a different baseline for what counts as no interaction.

A simple count example

Error counts provide a simple example of why the link function matters for interaction claims. Suppose that children make fewer errors as they get older. A Poisson model with a log link can represent this as a linear age effect on the log mean,

$$\log\{E(Y)\} = \beta_0 + \beta_1 \text{age},$$

with $\beta_1 < 0$. On the observed count scale, however, the same model implies a curved decreasing function, because $E(Y) = \exp(\beta_0 + \beta_1 \text{age})$. The curve is not an extra psychological mechanism. It is the consequence of mapping an additive model on the log scale back to non-negative counts (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972).

This becomes important when a second predictor is added. Imagine two groups that differ in overall error rate, but have the same age effect on the log scale. On the observed count scale, the absolute age-related reduction in errors will usually be larger where expected errors are high and smaller where expected errors are low. A Gaussian identity model fitted to the raw counts may then treat this changing absolute difference as evidence that age is moderated by group. A Poisson model with a log link can instead describe the same pattern as additive main effects on the log-count scale.

This does not mean that the Poisson-log model is always correct, or that observed-scale interactions are never meaningful. It means that the no-interaction baseline must be stated. For count outcomes, product-term coefficients and response-scale contrasts can answer different questions (Domingue et al., 2024; McCabe et al., 2022). The example shows that the central question is not only whether the outcome is Gaussian, Poisson, or binomial, but which scale defines the no-interaction baseline.

The practical consequence is that an interaction coefficient and a response-scale interaction are different quantities. A response-scale interaction can be defined as a change in an expected difference across levels of another variable. For example, for a binary or continuous predictor X and a moderator Z, one response-scale contrast is

$$\{E(Y \mid X = 1, Z = 1) - E(Y \mid X = 0, Z = 1)\} - \{E(Y \mid X = 1, Z = 0) - E(Y \mid X = 0, Z = 0)\}.$$

In a nonlinear GLM, this quantity depends on the inverse link, $g^{-1}(\eta)$. It can be nonzero even when the product-term coefficient is zero, and it can differ in size or sign from the product-term coefficient when the product term is nonzero. This is why product-term coefficients in GLMs should not be treated as automatic answers to response-scale moderation questions (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019).

The link-function problem is therefore a scale problem. To ask whether X and Z interact is to ask whether their effects depart from additivity on some scale. An identity-link model asks whether they depart from additivity on the observed mean scale. A binomial-logit model asks whether they depart from additivity on the log-odds scale. A binomial-probit model asks whether they depart from additivity on a probit scale. A chance-corrected link for forced-choice accuracy asks whether they depart from additivity on a scale that maps the linear predictor onto the interval from chance to perfect performance. These are not interchangeable scientific claims.

This point helps separate three problems that are often conflated. The first is a wrong-family problem: the outcome is analyzed with a distribution that poorly represents its support or sampling process, such as modeling trial counts as if they were unbounded Gaussian observations. The second is a wrong-link problem: the broad family may be plausible, but the link encodes an implausible scale of additivity, such as using a standard binomial-logit link for a task where performance has a known non-zero chance floor. Link misspecification can affect both fixed-effect inference in binary GLMs and link diagnostics in binary GLMMs (Czado & Santner, 1992; Yu & Huang, 2019). The third is a wrong-metric problem: the recorded outcome may itself be a distorted or coarsened measure of the construct, as with ordinal responses or some sum scores. These problems can occur together, but they are conceptually distinct.

The present paper focuses mainly on the second problem, especially when it affects interaction claims. This focus is narrower than the general warning that linear models can be inappropriate for non-Gaussian outcomes. Researchers may already choose a plausible family and still leave the scientific scale of additivity underspecified. For instance, choosing a binomial family for forced-choice accuracy acknowledges that the outcome is a number of successes out of a fixed number of trials. It does not by itself decide whether the mean should be linked to the linear predictor through a standard logit, a probit, a complementary log-log link, a flexible generalized logistic link, or a chance-corrected link that incorporates the lower asymptote of the task (Stukel, 1988).

A link-aware interaction analysis therefore requires more than selecting a software default. It requires asking four questions. First, what is the scientific quantity of interest: a coefficient on the link scale, a difference in probabilities, a risk ratio, a marginal effect, or another response-scale contrast? Second, what constraints does the outcome impose, such as bounds, discreteness, ordinal categories, skew, ceilings, floors, or chance performance? Third, what scale of additivity is scientifically plausible for the process under study? Fourth, does the substantive conclusion survive plausible alternative links? These questions do not guarantee a uniquely correct model, but they make the interaction claim explicit enough to evaluate.

This framework also clarifies what simulations in this paper are designed to show. The simulations do not ask whether every interaction in psychology is wrong. They ask whether an interaction can be generated or distorted when the fitted link defines a scale of additivity that differs from the scale used to generate the data. The answer matters because many psychological outcomes have exactly the properties that make link assumptions consequential: bounded accuracy, non-zero chance levels, ordinal ratings, counts, response times, and sum scores. The next sections develop these cases in turn.

Box 1. Prediction, Explanation, and the Scale of Interaction

The importance of link choice depends on the goal of the analysis. For purely predictive purposes, the primary criteria may be predictive accuracy, calibration, generalizability, and decision usefulness. A model can be useful if it predicts future observations well, even if its coefficients do not provide a faithful description of the psychological process that generated the data.

Scientific interaction claims require a different standard. When researchers claim that an effect is moderated, they usually claim more than that one model predicts slightly better than another. They claim that the effect of one variable changes across levels of another variable in a way that is meaningful for theory. That claim requires an estimand. It also requires a scale. A treatment effect might be larger in absolute probability points, larger in odds ratios, larger on a latent response scale, or larger only after a transformation. These are different claims, and they can lead to different conclusions.

This distinction is consistent with estimand-centered workflows that treat models as tools for generating clear target quantities rather than as collections of coefficients to be interpreted mechanically (Rohrer & Arel-Bundock, 2026). For interaction testing, the relevant target quantity should be stated before the product term is interpreted. If the target is a response-scale difference-in-differences, the analyst should compute and report that quantity. If the target is additivity on a latent or link scale, the link function should be justified as part of the scientific model.

The point is not that prediction and explanation are separate in practice. Predictive checks, calibration, and out-of-sample performance can all inform scientific modeling. The point is that good prediction alone does not specify the scale on which a moderation claim is being made. For scientific inference about interactions, the link function is part of the claim.

Forced-Choice Accuracy and the Non-Zero Chance Floor

Forced-choice accuracy is a useful case because it separates two modeling choices that are often conflated. The first choice concerns the sampling process. If a participant completes k_i trials and Y_i trials are correct, a binomial model is often appropriate:

$$Y_i \sim \text{Binomial}(k_i, p_i),$$

where p_i is the expected probability of a correct response. The second choice concerns the lower bound of meaningful performance. In a 2-alternative forced-choice task, or in a true/false task when random responding between the two alternatives is the relevant chance process, a participant who is guessing is expected to answer correctly on half of the trials. More generally, in an m-alternative forced-choice task, the chance level is $c = 1/m$. Thus, for many forced-choice tasks, the theoretically relevant response range is not simply 0 to 1, but chance to 1.

These two choices define three increasingly specific model specifications. A Gaussian identity model applied to proportions is the weakest specification. It treats accuracy as an approximately continuous response, ignores the binomial trial process, and does not respect the 0-to-1 bounds of accuracy. Such a model may be convenient for descriptive purposes, but it assumes additivity directly in observed accuracy units and can produce fitted values outside the possible range.

A standard binomial logit or probit model improves on this. It respects the binomial sampling process and constrains fitted probabilities to the interval from 0 to 1:

$$p_i = F(\eta_i),$$

where F is the logistic cumulative distribution function for a logit link, or the standard normal cumulative distribution function for a probit link. This model is right about the binomial structure of the data. However, it still assumes that the expected probability of a correct response can approach 0. In a task with a known chance level, this is a different assumption from the one implied by the response process. A standard binomial model is right about the sampling process, but wrong about the lower bound.

A chance-corrected binomial link keeps the binomial sampling model but maps the linear predictor to the interval from chance to 1:

$$p_i = c + (1 - c)F(\eta_i).$$

Here, c is the chance level, and F can again be a logistic or normal cumulative distribution function. For a 2-AFC or true/false task with chance level .50, this becomes

$$p_i = .50 + .50F(\eta_i).$$

This specification changes the no-interaction baseline. A product term in a standard binomial-logit or binomial-probit model tests additivity on a link scale defined over the full 0-to-1 probability range. A product term in a chance-corrected binomial model tests additivity on a scale that already encodes the task-implied lower asymptote. These are different scientific claims, even though both models use a binomial family.

The difference matters most when one group, condition, or region of the predictor space pushes performance close to chance. Suppose that age has the same effect in two groups on the latent performance-above-chance scale, with no true age-by-group interaction. On the observed accuracy scale, the lower-performing group may nevertheless show a compressed pattern near .50, because there is little meaningful room below chance. A standard logit or probit link places this lower compression near 0 rather than near .50. The fitted model can then use an age-by-group product term to absorb the systematic curvature created by the misplaced lower bound. The resulting interaction is not merely random Type I error. It is a pseudo-interaction produced by a mismatch between the scale used by the model and the scale implied by the task.

This is the point illustrated by Figure 3. The data were generated with no age-by-group interaction on the 2-AFC link scale. The identity-link model, the standard binomial-logit model, and the chance-corrected binomial-logit model nevertheless imply different fitted interaction patterns. The reason is not that the dataset contains multiple substantive truths. The reason is that each model defines a different baseline for no interaction. This is consistent with broader work showing that outcome-model misspecification can create false discoveries in interaction analyses (Domingue et al., 2024), and with work showing that product-term coefficients in nonlinear probability models are not, by themselves, interaction effects on the response scale (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019).

The chance-corrected link is not automatically the right model for every forced-choice dataset. It is most defensible when the chance level is known from the task design, guessing is a plausible lower-bound process, and performance below chance is not substantively meaningful for the target claim. Other cases may require richer models, for example if there are lapses, response biases, item heterogeneity, learning across trials, systematic below-chance performance, or participant-specific guessing strategies. Related work on binary GLMs and GLMMs also shows that link misspecification can affect likelihood-based inference, including in clustered binary data (Czado & Santner, 1992; Yu & Huang, 2019). The practical recommendation is therefore not to replace every standard binomial model with a chance-corrected model. It is to ask whether the link used in the interaction test encodes the scale implied by the task.

For scientific interaction claims, the key question is: on what scale should the age effect, group difference, treatment effect, or condition effect be additive? If the claim concerns absolute differences in percentage correct, an observed-scale estimand may be appropriate and should be reported directly as a response-scale contrast. If the claim concerns latent performance above chance, a chance-corrected link may be more appropriate. If the goal is prediction, standard and chance-corrected models can be compared by predictive accuracy, calibration, and generalizability. But when the conclusion is that one predictor moderates another, the link function is part of the claim (Loftus, 1978; McCabe et al., 2022).

Sum Scores as Bounded and Discrete Outcomes

Sum scores bring the link-function problem into the measurement model. Many psychological outcomes are observed as totals or averages across questionnaire items, task items, symptoms, errors, or ratings. These observed scores are often treated as if they were direct measurements of an approximately continuous psychological dimension. But the latent trait and the observed score are different objects. The latent trait may be approximately continuous and even roughly normal, whereas the observed score is produced by a limited number of discrete item responses, often through thresholds, endorsement probabilities, or success probabilities. The mapping from latent trait to observed total is therefore bounded, stepwise, and potentially nonlinear.

This distinction matters because a sum score can be nonnormal even when the underlying psychological dimension is not. With few items, ordinal categories, binary responses, unequal item thresholds, lower and upper bounds, and limited score ranges, the observed total may be skewed, compressed near floor or ceiling, or sparse in some regions. A fixed difference on the latent trait can correspond to different expected raw-score differences depending on where participants are located on the scale. Near the middle of the scale, a change in the latent trait may move responses across several item thresholds. Near the floor or ceiling, the same latent change may produce little visible change because many items are already at the minimum or maximum response. This is one reason why psychometric variables often deviate from simple normality (Micceri, 1989), and why treating ordinal responses as metric can produce false alarms, missed effects, reversals, and misleading interactions (Liddell & Kruschke, 2018).

This does not mean that sum scores are always wrong or scientifically useless. Sum scores can be descriptively useful, transparent, reliable, and adequate for many applied purposes. The key point is narrower: analyzing a sum score with a Gaussian identity model implies a strong scale assumption. It assumes that effects are additive in raw score units on the manifest total-score scale. In an interaction model, the no-interaction baseline is that the expected raw-score effect of one predictor is the same across levels of the other predictor. Sometimes this is exactly the estimand of interest. For example, a researcher or clinician may care about a two-point difference on a questionnaire because decisions are made on that reporting metric. In that case, the manifest-score interaction may be meaningful as a descriptive response-scale quantity.

However, many psychological interpretations go beyond the manifest score. Researchers often interpret a sum-score interaction as evidence that an effect differs on an underlying construct, such as anxiety, ability, symptom burden, motivation, or endorsement propensity. For that claim, additivity in raw score units may be a poor proxy for additivity on the latent or response-process scale. A group closer to the upper bound may show a smaller observed treatment effect not because the treatment is less effective, but because the score has less room to increase. Conversely, a group located near a dense region of item thresholds may show a larger raw-score change because the measurement scale is more sensitive there. The resulting product term can partly reflect the metric of the score rather than genuine moderation in the psychological process, a form of outcome-model or measurement-metric misspecification (Domingue et al., 2024).

A practical rule is to treat Gaussian identity models for sum scores as approximations whose plausibility depends on the score. They are more defensible when the score has many possible values, the distribution is roughly symmetric, observations are far from the lower and upper bounds, residual patterns are not strongly tied to fitted values, and the substantive claim is explicitly about raw score units. They are less defensible when the score has few levels, is strongly skewed, shows floor or ceiling concentration, contains ordinal item responses with unclear spacing, or is interpreted as a latent construct.

In those cases, researchers should consider models and sensitivity analyses that better reflect the response process. Cumulative-link or ordered-probit/logit models are useful when the outcome is ordered but not clearly interval-scaled (Agresti, 2010; Bürkner & Vuorre, 2019; Smith et al., 2020). Item-level ordinal or IRT-inspired models are useful when item thresholds, item difficulties, or differential item functioning are central to the claim. Binomial, beta-binomial, or Poisson-binomial approximations can be useful when the sum score counts endorsed symptoms, correct responses, or successes across items, especially when overdispersion or item heterogeneity is plausible. These models help because they preserve discreteness, bounds, and a nonidentity scale of additivity. They can also fail, however, if their assumptions are inappropriate, for example proportional odds, local independence, exchangeability of items, or stable item parameters.

The recommendation is therefore not to abandon sum scores. It is to report interaction claims in a way that makes the assumed scale visible. Researchers can inspect the score distribution, plot effects near bounds, compare Gaussian identity results with ordinal, bounded, or item-level alternatives, and report response-scale contrasts alongside model-scale parameters. If the interaction is stable across plausible specifications, the claim is stronger. If it appears only on the manifest sum-score metric, the conclusion should be framed as a property of that score, not automatically as moderation of the underlying psychological construct.

Within-Family Link Choice

The same issue can arise even when the outcome family is broadly appropriate. For binary, binomial, ordinal, count, or positive continuous outcomes, analysts often choose a familiar default link. In many cases, default links are reasonable. Logit and probit links, for example, often produce very similar fitted probabilities in the middle of the response range, especially after the usual approximate rescaling of coefficients. This practical similarity is real, and it explains why many applied conclusions are unchanged by switching between the two links. However, it should not be treated as a general guarantee, especially when inference depends on curvature rather than on average fitted probabilities. The link-misspecification literature shows that inadequate links can affect likelihood-based inference in binary GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019).

The choice between logit and probit is therefore not only a technical choice about coefficient scaling. A logit link is natural when the substantive interpretation is in terms of odds or odds ratios. In that case, additivity on the linear predictor scale means additivity in log odds, and exponentiated coefficients have a direct odds-ratio interpretation. A probit link is natural when the response process can be represented as a threshold crossing on an approximately normal latent variable, such as ability, signal strength, response tendency, or the latent propensity to give a correct response. This does not mean that probit is generally more correct than logit. It means that each link carries a different theoretical story about the scale on which effects combine.

For interaction claims, this distinction matters because interactions are especially sensitive to the shape of the link. Two links may give almost indistinguishable main-effect summaries in the central probability range but imply different compression near floors, ceilings, chance levels, or distributional tails. A product term in a logit model asks whether effects depart from additivity in log odds. A product term in a probit model asks whether they depart from additivity on a latent-normal threshold scale. These are closely related in many datasets, but they are not the same scientific null. Moreover, when the target claim concerns probabilities on the response scale, the product-term coefficient is not itself the interaction effect; response-scale contrasts or marginal effects are needed to state what changes in observed accuracy, endorsement probability, or risk (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019).

This is why logit-versus-probit sensitivity is more than a cosmetic robustness check. If a moderation claim appears only under one plausible link, or changes sign when the link changes, the result should not be described as a stable psychological interaction without qualification. Conversely, if the interaction conclusion is similar across logit, probit, and other theoretically defensible links, the claim is more credible. The question is not whether logit and probit always differ. They often do not. The question is whether the scientific moderation claim survives the set of plausible links for the outcome, task, and estimand.

This is also the sense in which choosing the correct family is necessary but not sufficient. A binomial family specifies a sampling model for successes out of trials. It does not by itself decide whether additivity is most plausible on a logit, probit, complementary log-log, flexible generalized-logistic, or chance-corrected scale (Stukel, 1988). In forced-choice tasks, the logit-versus-probit distinction is nested within a further link question: whether the lower asymptote should be 0, as in a standard binomial link, or the task's chance level, as in a chance-corrected link. An ordinal family likewise does not decide which cumulative link is best, and applied ordinal-regression work explicitly treats logit, probit, complementary log-log, negative log-log, and cauchit links as different modeling choices (Smith et al., 2020). A Gamma family does not by itself decide whether the mean is additive on the identity, log, or inverse scale. For interaction testing, these choices are not merely coefficient rescalings. They define different baselines for no interaction.

When link choice may matter less

The link-function problem does not imply that every link choice will change the substantive conclusion. In many applications, plausible links will lead to nearly identical fitted values and similar interaction summaries. This is most likely when predicted probabilities remain far from floors and ceilings, effects are small, and the candidate links produce almost overlapping curves over the range of the data actually observed. As Figure 5 suggests, logit and probit links can be practically very close in the central part of the probability range, even though they differ more clearly in the tails. Link choice is also less likely to threaten the conclusion when the interaction is reported as a response-scale quantity, such as a marginal effect or contrast, and when that quantity is robust across theoretically defensible links (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Finally, some theories do not depend on tail behavior. If the substantive claim concerns average differences in a well-observed central range, rather than saturation, guessing, floor effects, or ceiling effects, link sensitivity may be small. This is not a reason to ignore the link. It is a reason to report sensitivity analyses, so that readers can see whether the interaction claim is tied to a narrow modeling convention or survives reasonable alternatives.

Diagnostics Are Necessary but Not Sufficient

Diagnostics should be part of link-aware modeling, but they are not a complete solution to the link-function problem. The reason is a diagnostic paradox. A wrong link can induce a systematic pseudo-interaction that becomes statistically detectable as sample size increases, while the diagnostic tools used to identify the wrong link may still have limited power, or may flag only a vague form of misfit. In this situation, the interaction test can become sensitive to the artifact before the diagnostics become sensitive enough to clearly identify the source of the artifact.

Pregibon-style link tests address this problem more directly than generic model checks. They ask whether a simple expansion of the assumed link improves the fitted model (Pregibon, 1980). This is useful because the test is targeted at link adequacy rather than at residual misfit in general. Its limitation is equally important. A significant added-term test does not automatically tell the researcher which link is correct. It may reflect a wrong link, but it may also reflect a missing nonlinear predictor, an omitted covariate, a missing interaction, or another aspect of the systematic component. Conversely, a non-significant result does not certify that the chosen link defines the scientifically correct scale of additivity. It only means that this particular diagnostic did not detect this particular departure.

Simulation-based residual diagnostics provide a broader check. DHARMa implements randomized quantile residuals by simulating new responses from the fitted model and comparing the observed response with the fitted predictive distribution (Dunn & Smyth, 1996; Hartig, 2024). This makes residuals more interpretable for GLMs and GLMMs than raw, Pearson, or deviance residuals. In the worked example below, we use DHARMa to examine overall residual uniformity, dispersion, residual quantile patterns over fitted values, and residual quantile patterns over the focal predictor. These checks are useful because they can reveal broad misfit, overdispersion, zero-inflation or zero-deflation, and structured residual patterns. They are not, however, a certificate that the link function is theoretically adequate. A model may simulate data that look reasonably similar to the observed data while still defining additivity on the wrong scale for the interaction claim.

Model comparison tools, such as AIC, BIC, WAIC, or LOO, offer a third source of information. They can compare candidate models, provided that the candidate set is meaningful and specified before interpretation. Yet small differences in fit indices do not guarantee that an interaction is robust. Two links may have similar predictive performance and still imply different no-interaction baselines, especially near floors, ceilings, or chance levels. Conversely, a model with slightly better fit may not correspond to the estimand that the theory requires. For interaction claims, predictive adequacy and substantive adequacy are related but not identical.

For this reason, diagnostics should be treated as warnings, not certification. They can tell researchers that a model deserves scrutiny. They can sometimes reveal that a fitted model is plainly inadequate. But they usually do not distinguish, by themselves, between wrong link, missing nonlinearity, missing interaction, wrong family, and wrong measurement metric. The practical implication is that diagnostic checks should be paired with link sensitivity analyses and an explicit definition of the estimand. If the scientific claim concerns response-scale moderation, researchers should report response-scale contrasts or marginal effects and examine whether those quantities are stable across plausible links (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). If the claim concerns additivity on a latent or linear-predictor scale, the chosen link needs a theoretical justification. Diagnostics can support this argument, but they cannot replace it.

Simulation Studies

The previous sections established the conceptual problem: the link function defines the scale on which a no-interaction model is additive. The simulations ask how this conceptual problem translates into inferential risk in settings where the true data-generating process is known. Across simulations, the true product-term interaction is set to zero on the generating scale. We then fit models that impose different scales of additivity and record how often they detect a statistically significant interaction. The target is not to estimate how often published interactions are false. It is to quantify when wrong or inadequate links can generate pseudo-interactions under controlled conditions (Domingue et al., 2024; Rimpler et al., 2026).

This probability should not be interpreted as ordinary Type I error under a correctly specified null model. If a wrong link produces a systematic deviation from the correct no-interaction baseline, the fitted product term can partly absorb that deviation. In that case, increasing sample size does not necessarily protect the analysis. It can increase the probability of detecting the pseudo-interaction, making a wrong substantive conclusion more stable. The problem is therefore not only random false alarms at alpha. It is the power to detect a model-induced artifact. This point is conditional: it applies when the misspecified link creates a systematic response-surface difference that resembles an interaction in the fitted model.

The simulations quantify two quantities. First, for each substantive scenario, we estimate the probability of detecting an interaction when the generating model contains no product term on the relevant scale. Second, in a diagnostic worked example, we estimate how often model checks flag the misspecification that produced the pseudo-interaction. This comparison matters because link-oriented tests and simulation-based residual diagnostics can be useful, but they do not guarantee that the link is adequate for the interaction claim being tested (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

The simulation code is intentionally separated from the manuscript. The main text reports only compact summaries. The full data-generating processes, scripts, and extended output are reported in the Supplement and repository.

Simulation 1: Forced-Choice Accuracy

Simulation 1 extends the motivating 2-AFC example. We generated binomial accuracy data with a chance level of .50 and no age-by-group interaction on the chance-corrected logit scale. We then fit identity-link models to proportions, standard binomial-logit and binomial-probit models, and a chance-corrected binomial-logit model. The key outcome was the false-positive rate for the age-by-group product term.

The expected pattern is simple. When the fitted link matches the chance-corrected generating scale, the interaction test should be close to nominal. When the fitted model assumes a different scale of additivity, especially an identity scale or a standard binomial link that ignores the chance floor, the product term can become significant even though the generating interaction is zero.

Simulation 2: Sum Scores

Simulation 2 illustrates the measurement-metric version of the same problem. We generated a latent outcome with no true interaction, converted it to ordinal item responses through thresholds, and summed the items. We then tested the interaction on the observed sum-score scale. As an oracle comparison, we also tested the same interaction on the latent scale used to generate the item responses.

This simulation is not an argument against sum scores in general. It shows a narrower point: when the manifest score compresses the latent scale unevenly, additivity on the sum-score metric can differ from additivity on the latent metric. Interaction claims based on sum scores should therefore be accompanied by scale-aware sensitivity checks when the scientific claim concerns the underlying construct.

Simulation 3: Within-Family Link Choice

Simulation 3 asks whether link choice can matter even when the outcome family, experimental design, and random-effect structure are held fixed. We generated repeated binary responses in a two-group, two-condition design, with 15 trials per participant-condition cell and a subject random intercept. The true condition-by-group product term was zero on the generating link scale. We then crossed the generating link and fitted link, using logit and probit in both roles, and fitted all models as random-intercept GLMMs.

The purpose is not to show that logit and probit always disagree. They often do not, and they may be practically interchangeable when the data are concentrated in the middle of the response range. The narrower question is whether an interaction test can be affected by the curvature implied by a plausible alternative link, even when the family and random-effects structure are otherwise appropriate.

Because this simulation fits trial-level GLMMs, the additional descriptive quantities in the table are the median product-term coefficient and its standard error on the fitted link scale. They should not be read as response-scale changes in probability. Response-scale summaries would require a separate prediction-based estimand, which is not produced by this script.

Diagnostic Worked Example

Finally, we examined diagnostics in one selected scenario from Simulation 1. The data were generated from a chance-corrected binomial model with a .50 lower asymptote and no age-by-group product term on the generating link scale. We then fitted the misspecified standard binomial-logit model, which respects the binomial sampling process but assumes a lower asymptote of 0. The purpose was not to find the best possible diagnostic battery. It was to compare two rates in the same Monte Carlo setting: the false-positive rate for the age-by-group interaction under the wrong link, and the rate at which diagnostic checks flag the same wrong-link problem.

The primary diagnostic workflow used DHARMa residual checks for the fitted standard binomial-logit model (Hartig, 2024). In each replication, we simulated randomized quantile residuals and recorded four p-values: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual quantile patterns over age. We report these checks separately. We do not combine them into a single Holm-corrected or uncorrected omnibus decision, because they are partly overlapping summaries of the same simulated residuals and have different diagnostic meanings. We also report a Pregibon-style added-term link check as a secondary comparator, because it targets link adequacy more directly but only against a narrow added-term alternative

(Pregibon, 1980).

{#fig-diagnostic-worked-example}

The example illustrates the diagnostic paradox. The misspecified standard binomial link can produce a detectable interaction even though the generating model contains no age-by-group product term on the appropriate link scale. Some diagnostics may flag the problem, but they do not form one uniform signal, and none should be interpreted as a definitive test of link correctness. The Pregibon-style check is informative because it is link-oriented, but it still tests only a restricted alternative. The main lesson is therefore pragmatic: diagnostics are useful warnings, not certification. For interaction claims, they should be accompanied by a clear estimand, a justified link, and sensitivity analyses over plausible links.

A Minimal Workflow for Link-Aware Interaction Testing

The practical implication is not that researchers should stop testing interactions. It is that interaction claims should be made scale-aware. A minimal workflow is shown in Table 7.

The workflow is not a rule that can make interaction testing automatic. It is a way to put the main judgments in the open. Researchers still need theory to decide which contrast matters, which scale is meaningful, and which links are plausible. The table simply asks that these decisions be visible before the product term is interpreted. This matters because different quantities can be defensible for different questions: a linear-predictor coefficient may be useful for a model-based claim, whereas a response-scale contrast or marginal effect may be closer to the substantive claim made in words (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). Diagnostics and predictive checks can warn about misspecification, but they do not prove that the selected link is the true scientific scale (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980). Sensitivity analyses therefore play a pragmatic role. When the interaction conclusion is stable across plausible links, the claim becomes easier to defend. When it is not stable, the paper can still be informative, but the conclusion should be stated as conditional on a modeling choice. In this sense, link-aware reporting extends general quantitative-reporting principles: choices that affect the claim should be visible, justified, and interpretable (Mustillo et al., 2018).

Discussion

Interactions remain central to psychological theory, and the point of this paper is not that they are usually wrong. Some interaction claims will be robust across reasonable links, scales, and response-scale summaries. Others will not. Our claim is narrower: many interaction claims are underspecified because they do not state the scale on which additivity, and therefore no interaction, is being assumed. In GLMs, this scale is defined by the link function. This makes link choice a substantive modeling decision, not only a technical detail. The same issue may also extend beyond interactions to the broader study of nonlinear effects, because many psychological analyses search for curvature, thresholds, and changing slopes without first specifying the scale on which these patterns are expected to be absent or present (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022; Rohrer & Arslan, 2021; Wagenmakers et al., 2012).

The preregistered review showed why this matters for current practice. Interaction tests were common, but link functions were often implicit, and many interaction-testing articles used identity-link analyses in contexts where the outcome made the scale of additivity worth justifying. These findings do not show that the reviewed articles reached false conclusions. They show that the scale of many interaction claims is difficult to evaluate from the published report.

The conceptual framework separated three problems that are often mixed together: wrong family, wrong link, and wrong measurement metric. Forced-choice accuracy illustrates the wrong-link problem clearly. A standard binomial family respects trial counts, but a standard link may fail to encode the non-zero chance floor of the task. Sum scores illustrate the measurement-metric problem: an interaction on the manifest score scale may not be an interaction on the latent construct scale. Within-family links illustrate the more general point that plausible links can define different baselines for no interaction.

The simulations were designed to quantify pseudo-interactions, not to estimate the prevalence of false interactions in the literature. They show how link and metric choices can generate significant product terms even when the true interaction is zero on the generating scale. The diagnostic example shows a complementary point: diagnostics are useful, but they do not by themselves certify that a scientific interaction claim is made on the intended scale.

We deliberately reported the DHARMa checks separately rather than collapsing them into a single omnibus decision. In applied work, residual diagnostics are rarely equivalent to one clean hypothesis test. They are better understood as partially redundant probes of different possible symptoms of misspecification. This is precisely why passing one diagnostic, or even several diagnostics, should not be taken as strong evidence that an interaction claim is robust to link choice.

Several limitations are important. First, the review was descriptive and targeted to selected journals, not a census of psychological research. Second, we did not code whether published interactions were removable or nonremovable, because that distinction often cannot be recovered from published reports alone. Third, the simulations are intentionally simple. More realistic data may include random effects, item heterogeneity, lapses, response bias, nonlinear predictors, and missing data. Fourth, chance-corrected links are appropriate only when a chance floor is theoretically justified. They are not a universal replacement for standard binomial models. Finally, logit and probit links often lead to very similar conclusions, especially away from boundaries; the paper's claim is that link choice can matter for interaction claims, not that it always does.

The main contribution is therefore practical. Link-aware interaction testing does not make moderation claims automatic, and it does not identify one universally correct scale. It asks researchers to make the main judgments visible: what outcome family is plausible, what link defines the no-interaction baseline, what response-scale quantity answers the substantive question, and whether the conclusion survives plausible alternatives. When the same conclusion holds across defensible links, the interaction claim is strengthened. When it changes, the result can still be informative, but it should be reported as conditional on a modeling choice. This workflow complements existing guidance on nonlinear interactions, marginal effects, and prediction-based interpretation (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). In GLMs, the link function is not an afterthought. It is the scale on which no interaction is defined.

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Table 1

Descriptive summary of the preregistered review of interaction testing and link-function reporting.

Quantity	n	Denominator	Percent	95% CI
Eligible empirical articles reviewed	331	331	100.0%	[98.9%, 100.0%]
Articles testing at least one interaction	200	331	60.4%	[55.1%, 65.5%]
Interaction-testing articles using a non-identity link	58	200	29.0%	[23.2%, 35.6%]
Interaction-testing articles explicitly reporting the link function	46	200	23.0%	[17.7%, 29.3%]
Interaction-testing articles with a potentially problematic identity-link analysis	144	200	72.0%	[65.4%, 77.8%]
Potentially problematic identity-link cases reporting at least one significant interaction	124	144	86.1%	[79.5%, 90.8%]

Table 2

Broad response-variable categories in interaction-testing articles. Categories are non-exclusive, so one article can contribute to more than one row.

Outcome type	n	Denominator	Percent
Binary / accuracy / proportions	86	200	43.0%
Sum scores / composites	77	200	38.5%
Ordinal / Likert / ratings	49	200	24.5%
Response times / durations	29	200	14.5%
Neural / physiological measures	23	200	11.5%
Difference / distance scores	21	200	10.5%
Counts / error counts	12	200	6.0%
Correlations / associations	8	200	4.0%

Table 3

Simulation 1: false-positive interaction rates and median model-implied changes in the group gap for forced-choice accuracy data. CI = Wilson 95% confidence interval.

Scenario	Model	Fits	Sig	False-positive rate	95% CI	Median gap change
Middle range	Chance-corrected logit	300	17	0.057	[0.036, 0.089]	-4.249
Middle range	Identity	300	300	1.000	[0.987, 1.000]	-6.751
Middle range	Standard logit	300	300	1.000	[0.987, 1.000]	-6.733
Middle range	Standard probit	300	300	1.000	[0.987, 1.000]	-6.638
Near ceiling	Chance-corrected logit	300	13	0.043	[0.025, 0.073]	3.746
Near ceiling	Identity	300	300	1.000	[0.987, 1.000]	6.090
Near ceiling	Standard logit	300	300	1.000	[0.987, 1.000]	4.571
Near ceiling	Standard probit	300	300	1.000	[0.987, 1.000]	5.317
Near chance floor	Chance-corrected logit	8	1	0.125	[0.022, 0.471]	0.000
Near chance floor	Identity	300	300	1.000	[0.987, 1.000]	-10.552
Near chance floor	Standard logit	300	300	1.000	[0.987, 1.000]	-10.325
Near chance floor	Standard probit	300	300	1.000	[0.987, 1.000]	-10.458

Table 4

Simulation 2: false-positive interaction rates and median model-implied changes in the group gap for observed sum scores versus the latent generating scale. CI = Wilson 95% confidence interval.

Scenario	Model	Scale	Fits	Sig	False-positive rate	95% CI	Median gap change
High scores	Bounded-score logit	sum score	300	109	0.363	[0.311, 0.419]	1.609
High scores	Latent oracle	latent theta	300	16	0.053	[0.033, 0.085]	0.003
High scores	Sum-score identity	sum score	300	112	0.373	[0.321, 0.429]	1.328
Low scores	Bounded-score logit	sum score	300	92	0.307	[0.257, 0.361]	-1.988
Low scores	Latent oracle	latent theta	300	18	0.060	[0.038, 0.093]	0.011
Low scores	Sum-score identity	sum score	300	216	0.720	[0.667, 0.768]	-1.762
Middle scores	Bounded-score logit	sum score	300	123	0.410	[0.356, 0.466]	-0.700
Middle scores	Latent oracle	latent theta	300	16	0.053	[0.033, 0.085]	0.018
Middle scores	Sum-score identity	sum score	300	29	0.097	[0.068, 0.135]	-0.547

Table 5

Simulation 3: false-positive interaction rates for the condition-by-group product term in repeated-trial random-intercept GLMMs. CI = Wilson 95% confidence interval.

Generating link	Fitted link	Link status	Fits	Sig	False-positive rate	95% CI	Median beta interaction	Median SE
logit	logit	Matched link	300	19	0.063	[0.041, 0.097]	-0.003	0.067
logit	probit	Wrong link	300	20	0.067	[0.044, 0.101]	-0.011	0.039
probit	logit	Wrong link	300	41	0.137	[0.102, 0.180]	0.071	0.076
probit	probit	Matched link	300	15	0.050	[0.031, 0.081]	-0.003	0.043

Table 6

Diagnostic worked example: false-positive interaction rate and separate diagnostic detection rates in one selected scenario.

Quantity	Fits	Sig	Rate	95% CI
Wrong-link interaction	300	255	0.850	[0.805, 0.886]
DHARMa uniformity	300	0	0.000	[0.000, 0.013]
DHARMa dispersion	300	0	0.000	[0.000, 0.013]
DHARMa residual quantiles over fitted values	300	44	0.147	[0.111, 0.191]
DHARMa residual quantiles over age	300	90	0.300	[0.251, 0.354]
Pregibon-style added-term link check, secondary	300	179	0.597	[0.540, 0.651]

Table 7

A minimal workflow for link-aware interaction testing.

Step	Question	Why it matters for interactions	Minimal reporting
Define the interaction estimand	What quantity would answer the moderation question?	Product terms, marginal effects, and difference-in-differences can answer different questions.	State the target contrast, scale, moderator values, and whether the estimand is descriptive, associational, or causal.
Inspect outcome constraints	What restrictions does the outcome impose?	Bounds, discreteness, floors, ceilings, and chance levels can make equal observed differences misleading.	Report the response range, discreteness, trial or item count, chance level, and visible floor or ceiling effects.
Justify the outcome family	What sampling model and support are plausible?	A wrong family can create residual structure and pseudo-interactions before link choice is considered.	Name the family and justify it from the outcome type and design, such as trials, counts, positive times, or ordered categories.
Justify the link	On which scale should effects add if there is no interaction?	The link defines the no-interaction baseline. A wrong link can turn curvature into a product term.	Name the link and give the substantive or design reason, such as odds, latent threshold, log mean, or chance floor.
Distinguish linear-predictor interaction from response-scale interaction	Is the claim about the model scale, the observed scale, or both?	A significant product coefficient may not correspond to meaningful response-scale moderation.	Report the product coefficient when relevant, and also predicted means, contrasts, marginal effects, or difference-in-differences.

Figure 1

Same family, different additivity scale. Both panels are compatible with positive response-time outcomes, but the no-interaction baseline differs. A Gamma log-link model assumes additivity in log means, which implies proportional change on the millisecond scale. A Gamma identity-link model assumes additivity directly in milliseconds.

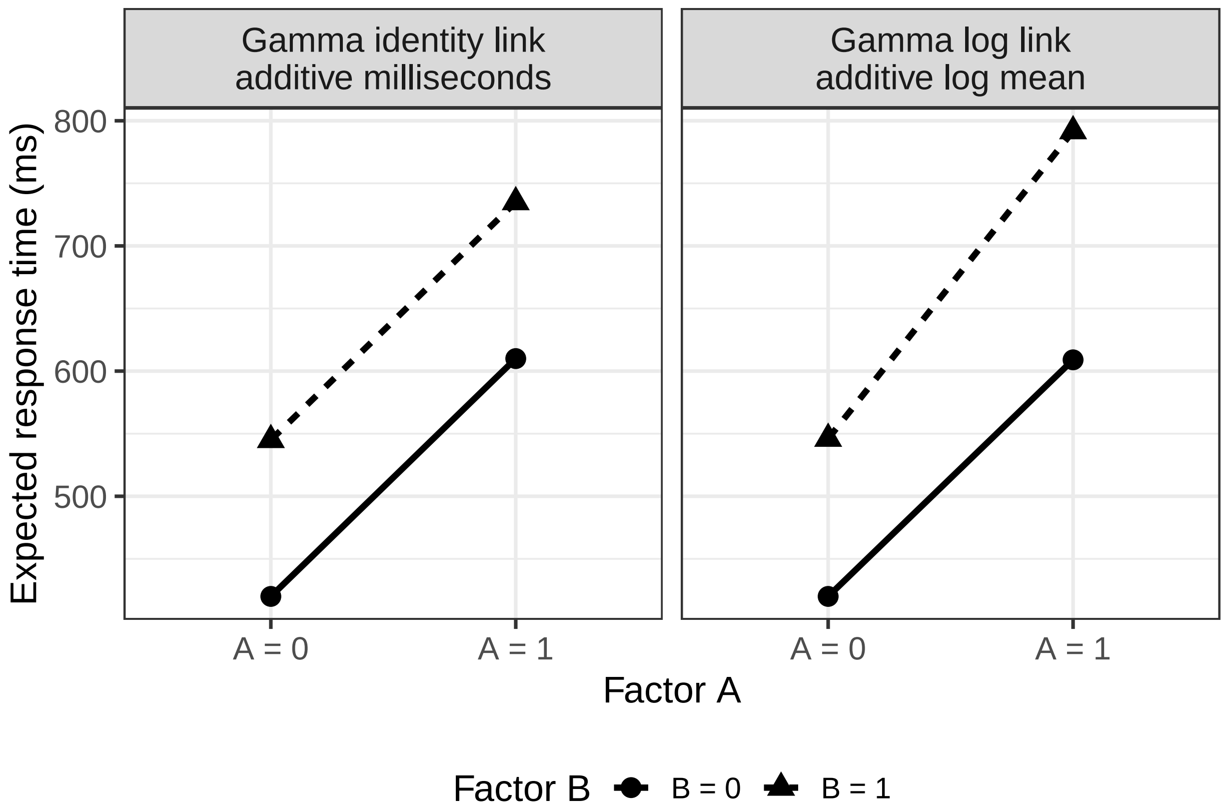

Figure 2

Same outcome family, different link functions. Both curves are compatible with binomial outcomes, but they encode different no-interaction scales: the standard logit link maps the linear predictor to the full 0-to-1 probability range, whereas the chance-corrected link maps it to the .50-to-1 range expected in a two-alternative forced-choice task.

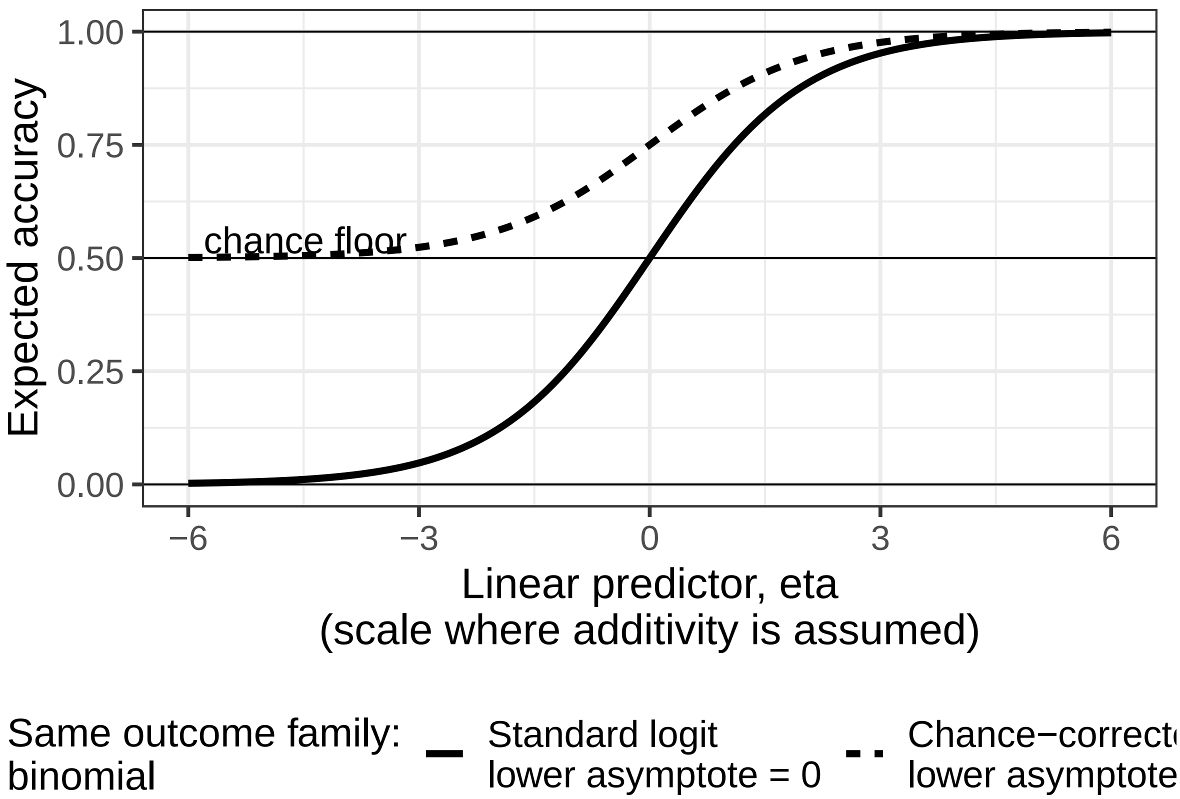

Figure 3

Same data, different no-interaction baselines. Data were simulated from a 2-alternative forced-choice accuracy task with 50 trials per participant and a .50 chance floor. The data-generating model included age and group effects, but no age-by-group interaction on the chance-corrected 2-AFC logit scale. Panel A shows the true response-scale pattern implied by that model. Panel B shows one simulated dataset. Panel C fits the same data with three models. The Gaussian identity model is simple and interpretable on the observed accuracy scale, but it can produce predictions outside the 0-to-1 range and ignores the binomial sampling process. The standard binomial logit/probit model, shown here with a logit link, respects the binomial process and the 0-to-1 range, but assumes a lower asymptote at 0. The chance-corrected binomial link incorporates the task's chance floor. The question is not only which curve fits, but which scale defines the scientific null hypothesis of no moderation.

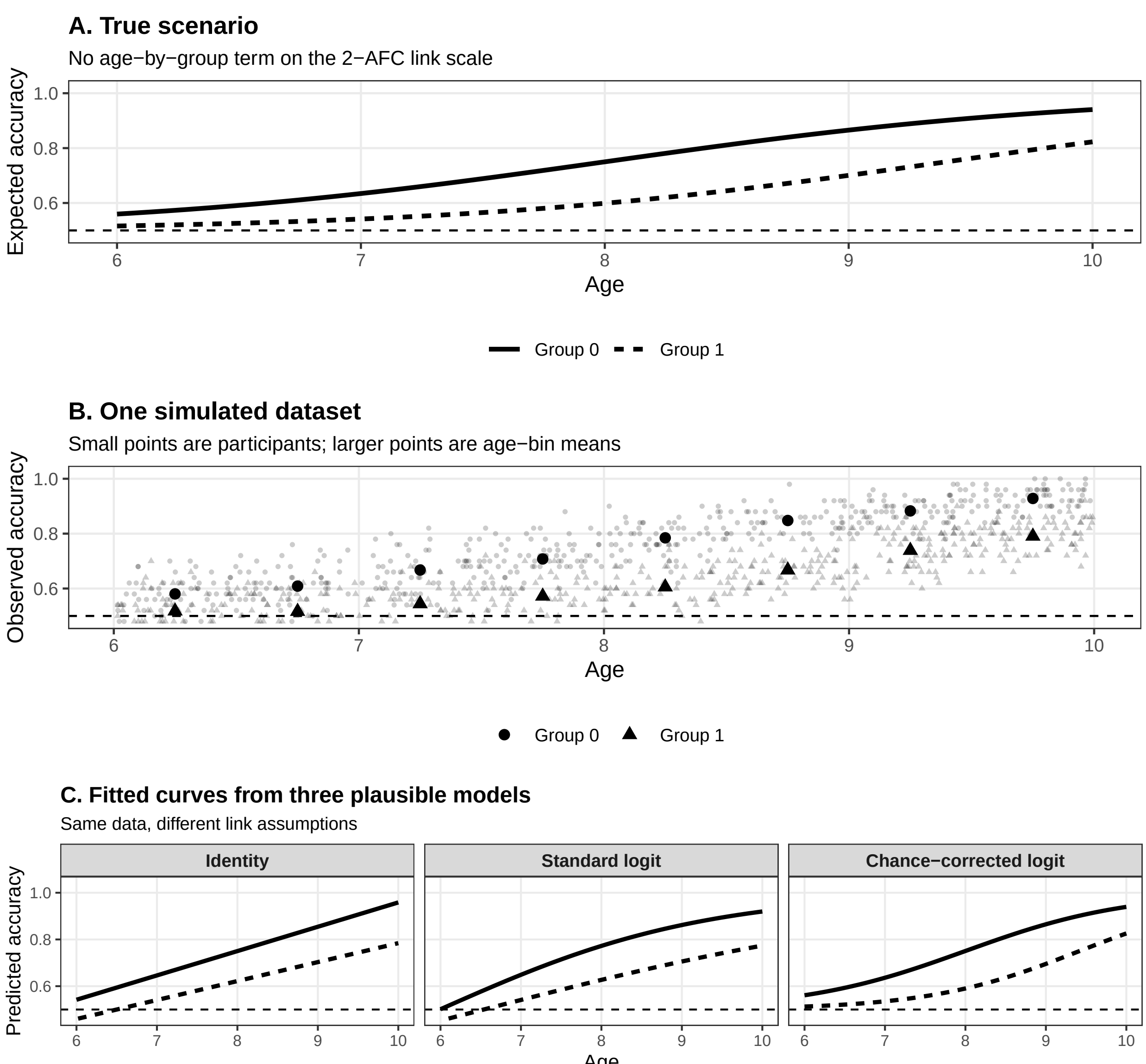

Figure 4

A didactic count example. The two groups have the same age effect on the log-count scale and no age-by-group product term on that scale. After the inverse log link, the expected error counts decrease nonlinearly on the observed scale, and absolute group differences change with age.

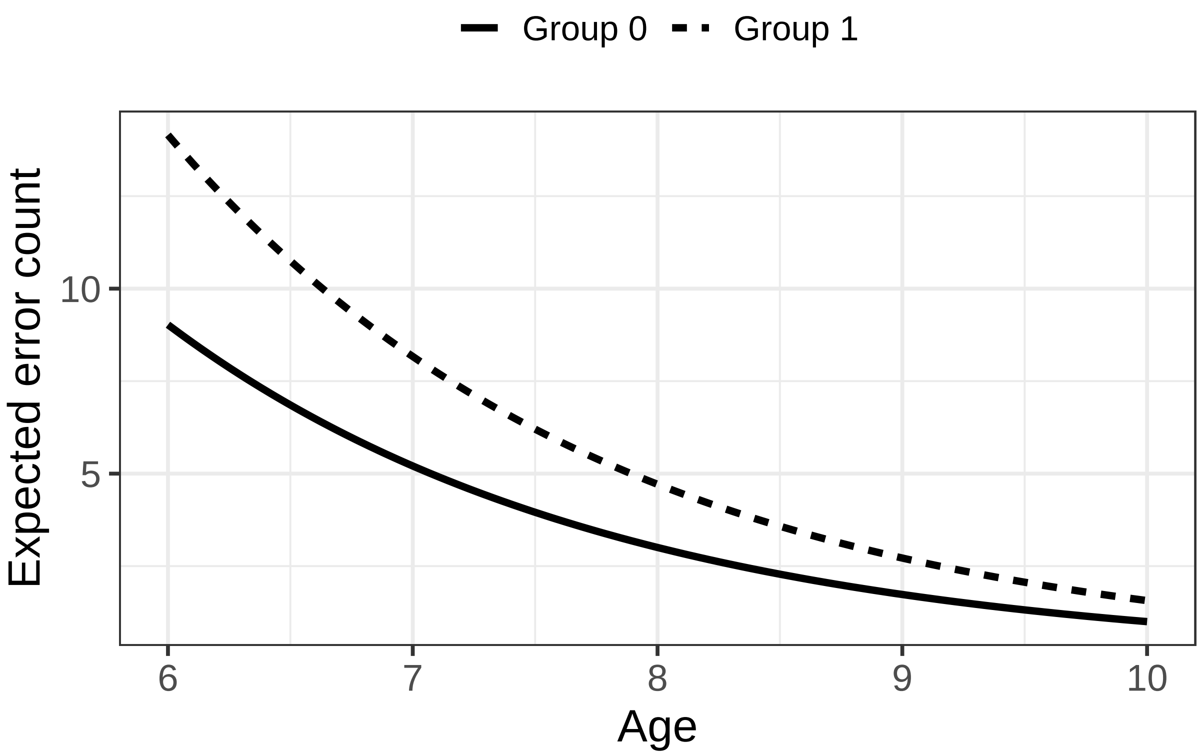

Figure 5

Logit and probit links can be very similar in the middle of the probability range after approximate rescaling, but they are not identical in the tails. For interaction testing, these tail differences matter because floors and ceilings change the amount of response-scale room left for an effect.

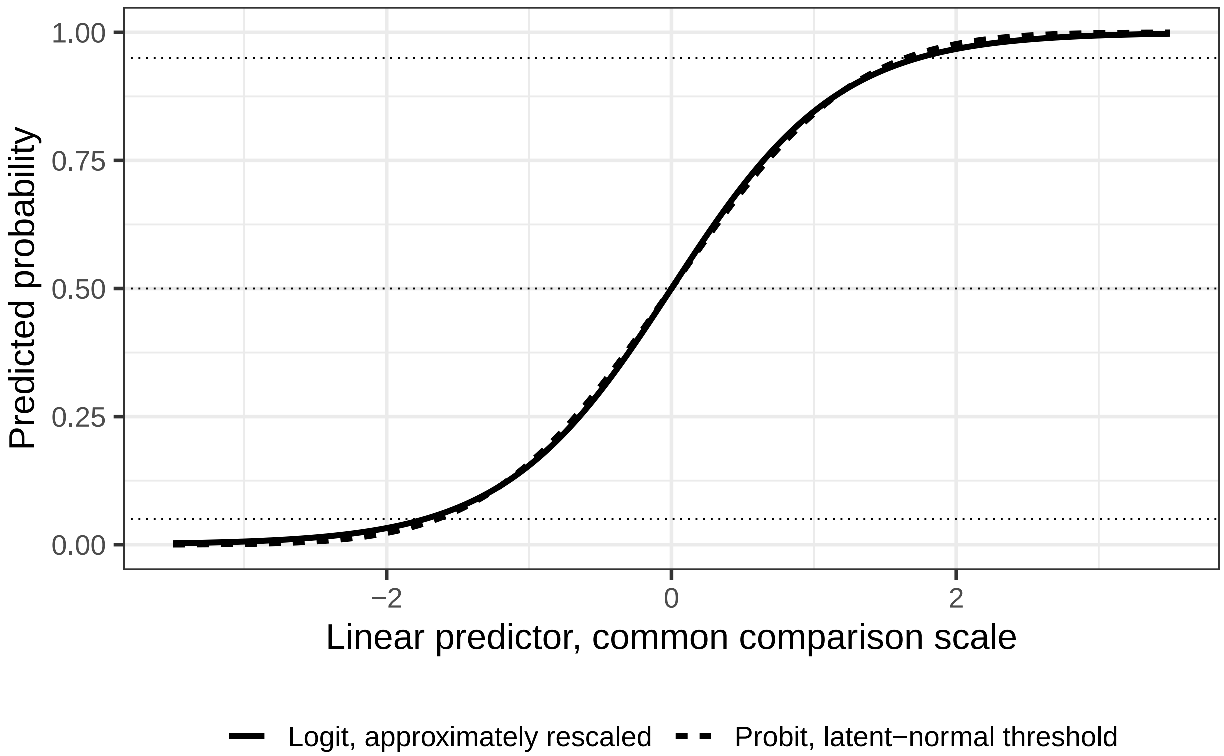

Figure 6

Simulation 1: scenario visualization and model consequences for forced-choice accuracy. The panels show the deterministic scenario implied by the data-generating model, the corresponding observed-scale group gap across age, and false-positive interaction rates across repeated simulations. Data are generated with no age-by-group interaction on the chance-corrected scale.

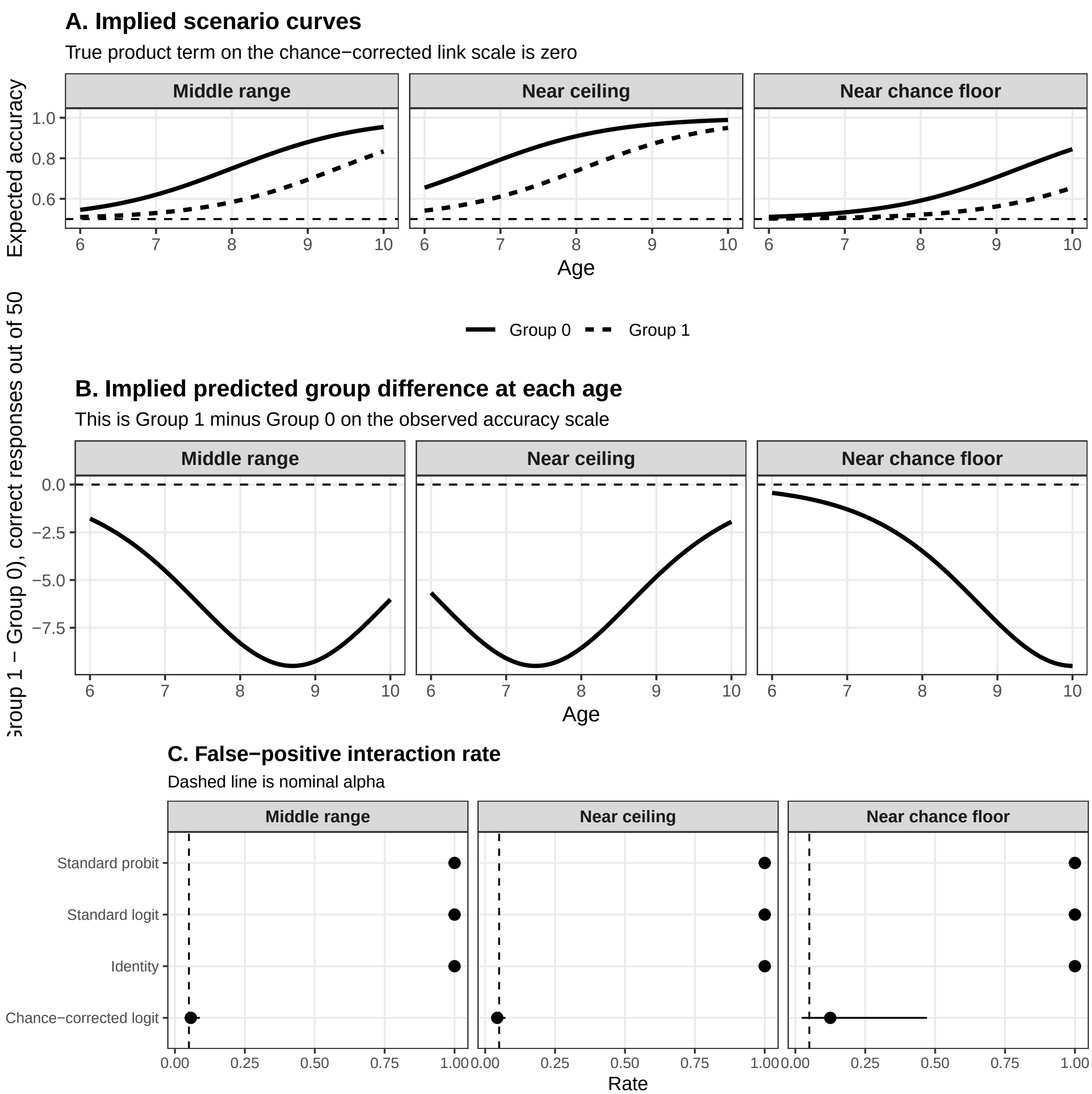

Figure 7

Simulation 2: scenario visualization and model consequences for sum scores. The panels show how the latent data-generating model is mapped onto the bounded sum-score scale, how the predicted group difference changes across the predictor range, and false-positive interaction rates across repeated simulations.

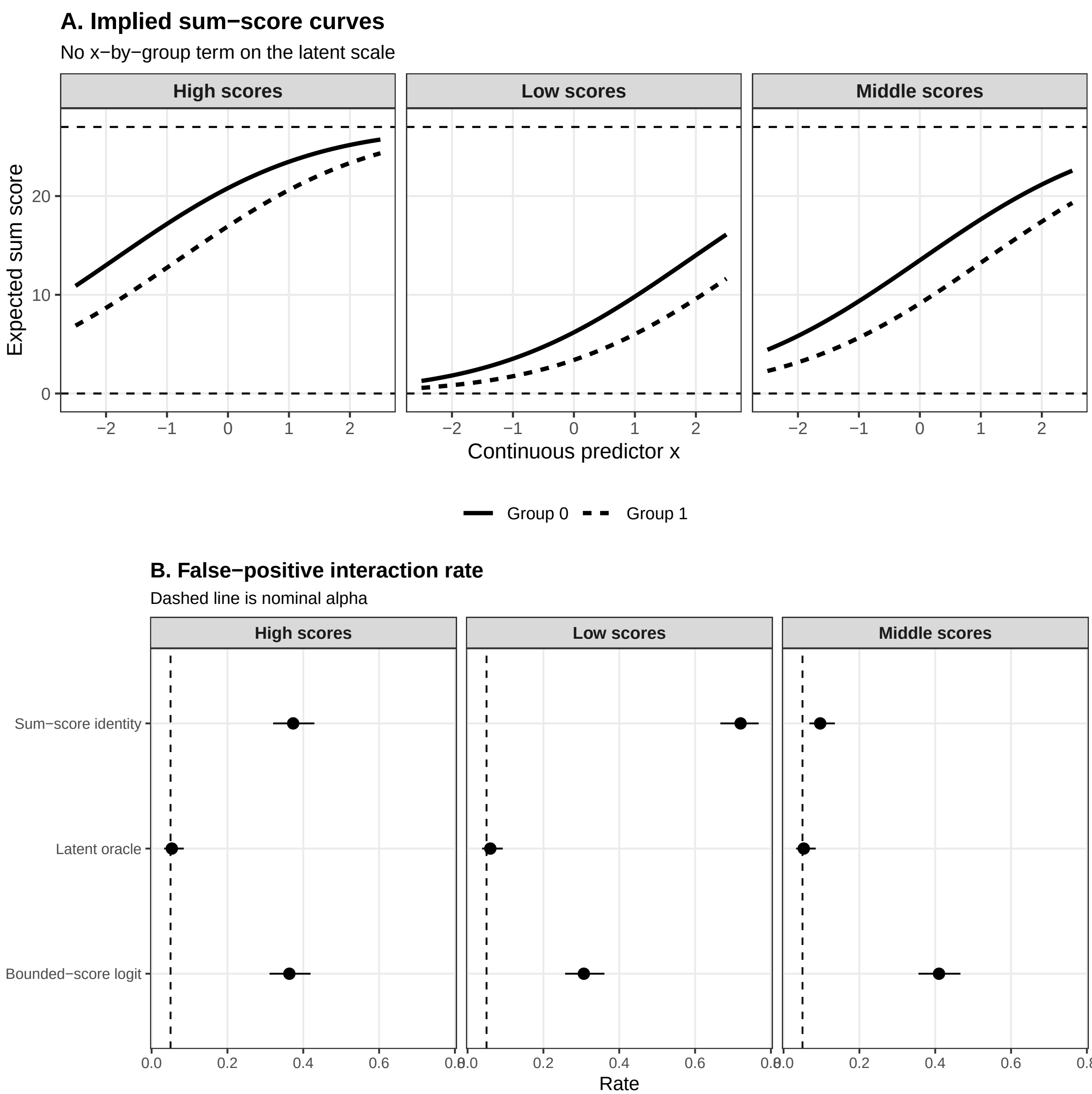

Figure 8

Simulation 3: false-positive interaction rates in repeated-trial random-intercept GLMMs. Data were generated under logit or probit links with no condition-by-group product term on the generating scale, then fitted using logit and probit links. Error bars are Wilson 95% confidence intervals; the dashed line marks nominal alpha.